Project ID: 26-IT-56

Project Proposal

Smart Emergency Traffic Light Control System

SURAWEERA R.B.J.H
IT22298676

T.A.D.T JAYAMPATHI
IT22294234

SIRIWARDANA W.L.N.S
IT22070494

KITHSANDU E G H
IT22606242

Introduce The Problem

Problem & Need

Fixed-Time Signal Limitations

Traditional traffic lights operate on pre-set timers and cannot adapt to real-time traffic density or detect approaching emergency vehicles, causing unnecessary stops.

Critical Response Delays

Ambulances and fire brigades face avoidable delays at busy intersections, losing precious minutes that are crucial for patient survival and fire control.

Increased Safety Risks

Unexpected delays and forced maneuvers at red lights increase risks to life, property, and public safety for both emergency responders and general commuters.

Sri Lankan Context

95% Passenger traffic depends heavily on road transport networks

11:32 Average ambulance response time in Colombo limits (mins:secs)

14:49 Island-wide average response time for 1990 Suwa Seriya

Sources: RDA National Road Master Plan 2021; 1990 Suwa Seriya Foundation Reports

Users and Stakeholders

The group of people and organizations involved in the Smart Emergency Traffic Light Control System.



Core Operators

Direct users who interact with the system to initiate emergency requests and require immediate route clearance.

- Ambulance Drivers
- Fire Brigade Drivers
- Emergency Dispatchers



Beneficiaries & Admins

Entities that benefit from improved response times or manage the infrastructure and data flow.

- Critical Patients
- Hospitals & Clinics
- Traffic Authorities
- System Administrators



General Public

Road users who experience safer intersections and reduced congestion during emergency events.

- Private Vehicle Drivers
- Public Transport Users
- Pedestrians

Existing Solutions & Their Limitations



What exists today

- Some traffic systems use:
- fixed timer traffic signals
- limited sensor based monitoring
- camera based traffic observation
- emergency vehicle tracking only
- simple emergency preemption systems

Main limitations

- no full integration
- no direct mobile app to traffic signal communication
- no route based pre clearing
- no strong support for multiple emergency vehicles
- no smart warning support in difficult traffic situations

Impact on Users & Society

Impact on users

- emergency vehicles get a clearer path
- drivers receive better support during urgent situations
- delays at junctions are reduced
- the system becomes more responsive to real traffic conditions

Impact on society

- better road safety
- improved emergency healthcare response
- better rescue response during fires and accidents
- improved smart city traffic management







Comparison: Traditional vs. Smart Traffic Systems

A side-by-side look at how our proposed smart system dramatically improves upon current traditional traffic management.

Feature	Traditional Traffic Systems	Proposed System
Traffic monitoring	Limited	Live camera based monitoring
Emergency tracking	Not supported	GPS based real-time tracking
Signal control	Fixed timer	Intelligent adaptive control
Traffic clearing	Reactive	ETA-based pre-clearing
Emergency warning	Not available	Siren-assisted warning
Multi-vehicle support	Not supported	Priority-based sequencing

Our system is more complete because it combines real-time monitoring, precise tracking, intelligent decision-making, and dynamic control within one integrated platform.

Solution Overview

1

Driver Activates App

Emergency driver enables "Emergency ON" mode. App sends GPS, speed, destination & priority to cloud.

2

Density Detection

4-way cameras & Edge AI estimate real-time traffic density at the junction and push data to cloud.

3

Cloud Decision

Cloud fuses ETA & density data. Selects strategy: Single EV Priority vs. Multi-EV Conflict Management.

4

Action & Alert

Traffic lights pre-clear the route. Buzzer activates if congestion remains critical or unmanageable.

5

Fail-Safe Mode

System automatically switches to "Yellow Blinking" mode upon detecting any technical fault or cyberattack.



01
Project Overview



IT22298676 | SURAWEERA R.B.J.H.

GPS-Based Emergency Vehicle Detection

Contribution

- Creating the mobile app for the emergency drivers
- Real-time GPS tracking & speed monitoring
- Role-based UI (Ambulance vs. Fire Brigade)
- Patient priority selection (High/Medium)

Novelty of My Component

- **Smart ETA Calculation:** It calculates the exact arrival time directly inside the phone, not just the basic location.
- **Priority Selection:** The driver can manually choose the patient's emergency level (High or Medium priority).
- **Set Destination**
- **High Security:** It uses data encryption to send live tracking data safely to the cloud, stopping any hackers.

GPS-Based Emergency Vehicle Detection

Problems Solved by GPS Tracking System

- There is no real-time communication channel from emergency vehicle drivers to traffic systems.
- Distance to upcoming junctions is not automatically calculated.
- Estimated Time of Arrival (ETA) information is not continuously updated.
- Critical route information is not directly shared with traffic control systems.

Research Objectives

Collect real-time GPS location from the emergency vehicle.

Capture current speed and acceleration data.

Calculate distance to the target junction automatically.

Estimate arrival time (ETA) at critical intersections.

Send all collected data securely to the cloud platform.

Support route and destination-based traffic preparation.

Methodology & Technologies

Research Domain & Approach

Methodology:

System Development & Evaluation (Design, Develop, Test)

Key Domains:

Mobile Application Development, Cloud Computing, Smart Transport (ITS)

Core Technologies & Tools

- **Flutter Mobile Dev** — Cross-platform App
- **Firebase Cloud DB** — Real-time Data Sync
- **Google Maps API** — Route & ETA Calculation
- **AES-256 & SHA-256** — Encryption & Hashing

1. Requirement Analysis
2. App Design & Logic
3. Development & Integration
4. Testing & Validation

Distance Calculation Logic

Haversine Formula

$$a = \sin^2(\Delta\varphi/2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda/2)$$
$$c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1-a)})$$
$$d = R \cdot c$$

Used to calculate the precise distance between the moving emergency vehicle (GPS) and the target traffic junction coordinates to trigger pre-clearing logic.

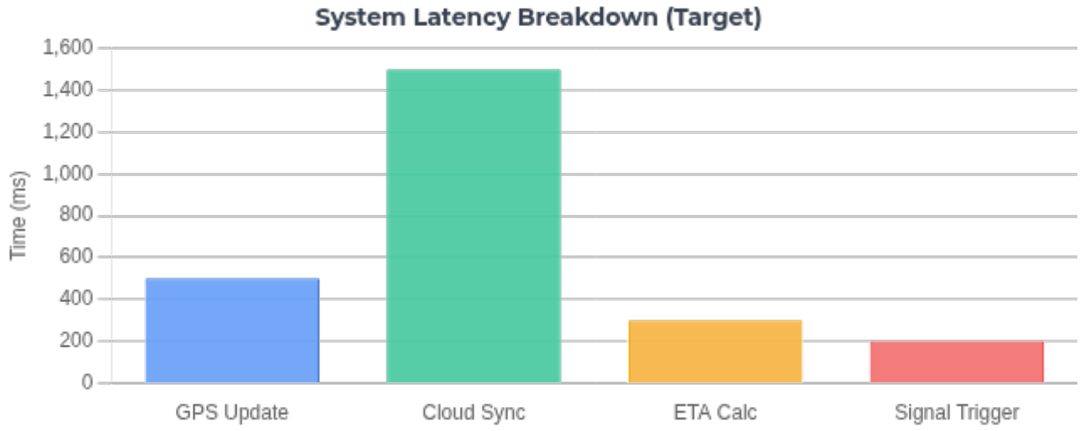
Key Metric

ETA < 3s Latency

Validation, Security & Ethics

Performance Validation Metrics

< 3s Cloud Latency Target	**98%** Message Delivery Rate
±5m GPS Accuracy	**< 200ms** App Response Time



Data Security Measures

- Secures live GPS, ETA, and priority data sent from the app to the cloud, preventing hackers from intercepting it
- Ensures only verified drivers can log in to the app and uses hashing to confirm the data is not changed.
- Only authorized emergency drivers can log in, activate the 'Emergency ON' mode, and set the priority level of the situation.

Ethical Considerations

We strictly protect user privacy. The app tracks the location *only* when the Emergency ON' button is pressed. Also, we do not save any patient details; the app only sends the emergency level (High or Medium) to the system.

Implementation Overview

Mobile App Interface Flow



Driver selects vehicle type & patient priority -> App switches to tracking mode

Key Dependencies & Technologies

- Maps & Directions API
- Firebase Cloud DB
- GPS Location Services
- AES-256 Encryption

System Output Data Flow

Vehicle App → Cloud API → Control Logic

GPS Lat/Long | Speed | ETA | Priority Level

The app should be simple so emergency drivers can use it quickly under pressure.

Workflow: How It Works

Our system activates through a simple, yet robust, workflow ensuring a rapid and coordinated response for emergency vehicles.

Driver Activation
Emergency mode is initiated via a dedicated mobile app.

Real-time Tracking
The app continuously transmits the vehicle's precise location.

Data Collection
Current speed, acceleration, and route data are gathered.

Destination Input
The emergency vehicle's destination is specified in the app.

Cloud Transmission
All collected data is securely sent to the central cloud platform.

ETA Calculation
The system dynamically calculates distances and estimated arrival times.

System Update
Traffic control systems receive updated emergency vehicle information.

This seamless process ensures the wider traffic management system is fully informed, enabling it to proactively clear the route ahead of the emergency vehicle's arrival.



IT22294234 |T.A.D.T JAYAMPATHI

Camera Based Traffic Density Detection

Bachelor of Science (Hons) in Information Technology Specializing in Information Technology

Camera Based Traffic Density Detection

Contribution

Real-time traffic density monitoring system using affordable camera hardware and edge computing for 4-way junctions.

- **4-Direction Monitoring**: Uses cameras on all approaches.
- **Edge AI Processing**: Raspberry Pi-based vehicle counting.(data processing)
- **Thermal Support:** Improves reliability in poor visibility.

Novelty

- Integrates thermal support logic for night/rain conditions often missed by standard cameras.

.

Research Objectives

- **Capture Live Video:** Acquire real-time feeds from 4 cameras covering all junction approaches.
- **Detect & Count Vehicles**: Implement computer vision models on Raspberry Pi to identify cars, buses, and trucks.
- **Estimate Traffic Density**: Calculate congestion levels per road and send processed data to the cloud.
- **Enhance Reliability:** Explore thermal support methods to maintain accuracy during low-light or adverse weather conditions.

Problem Definition

Traditional traffic systems at junctions lack an **accurate vehicle counting system** and rely on **fixed timer signals instead of real-time traffic data manly poor visibility conditions such as night, rain, or fog reduce camera detection accuracy**, making it difficult to understand the actual traffic situation.

Problems solved

- Use **thermal support approach** to improve detection during poor visibility
- Calculate **traffic density for each road direction** automatically using four cameras

Main need

The system must know which road is crowded and which road is clear

Domain Knowledge & Technologies

Domain Knowledge

This component belongs mainly to Real-Time IoT & Embedded Systems, with support from Computer Vision and AI.
Computer Vision helps the system detect and identify vehicles from camera images and videos to estimate traffic density at the junction



Technologies

- Four cameras
- Raspberry Pi
- Python
- YOLO
- Firebase Cloud
- CNN (Convolutional Neural Network)
- GANs (Generative Adversarial Networks)

Validation, Security & Ethics

Validation

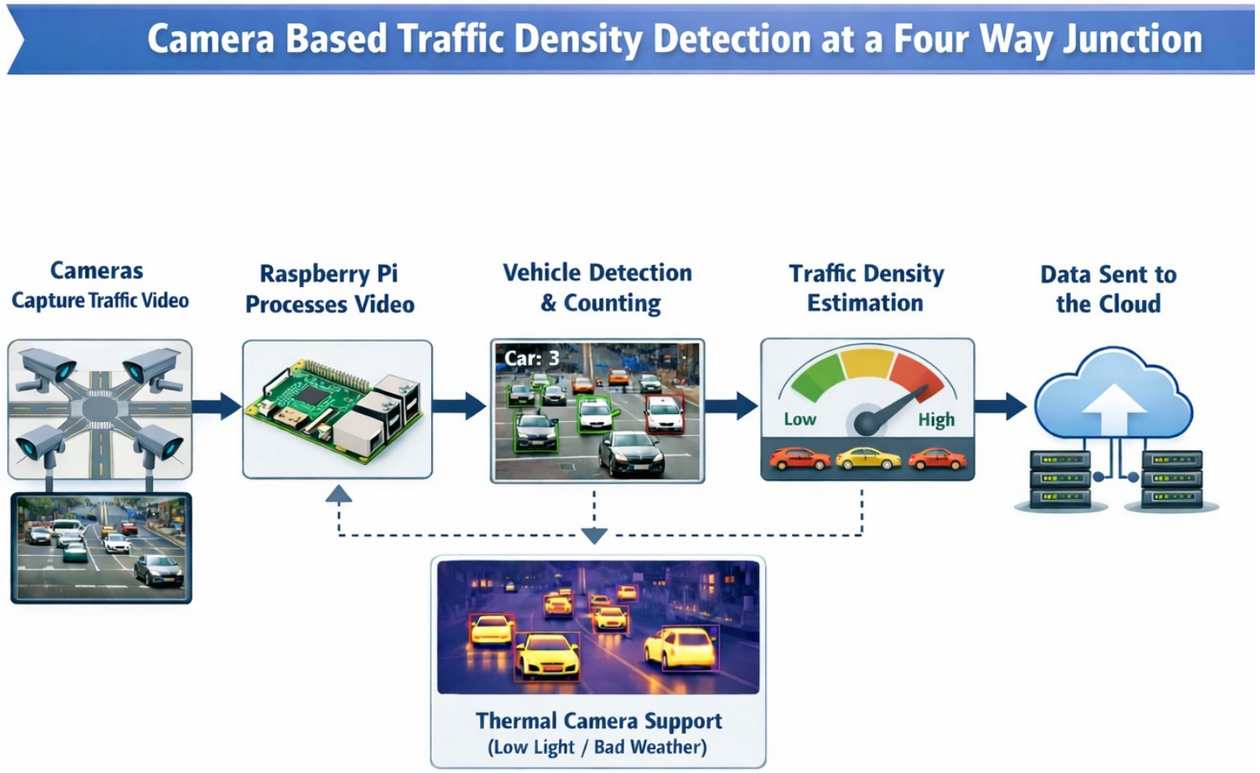
1. System is tested using **vehicle detection and counting accuracy**
2. Checked in **day, night, rain, and fog conditions**
3. **Fast response** with low delay using edge processing

Security

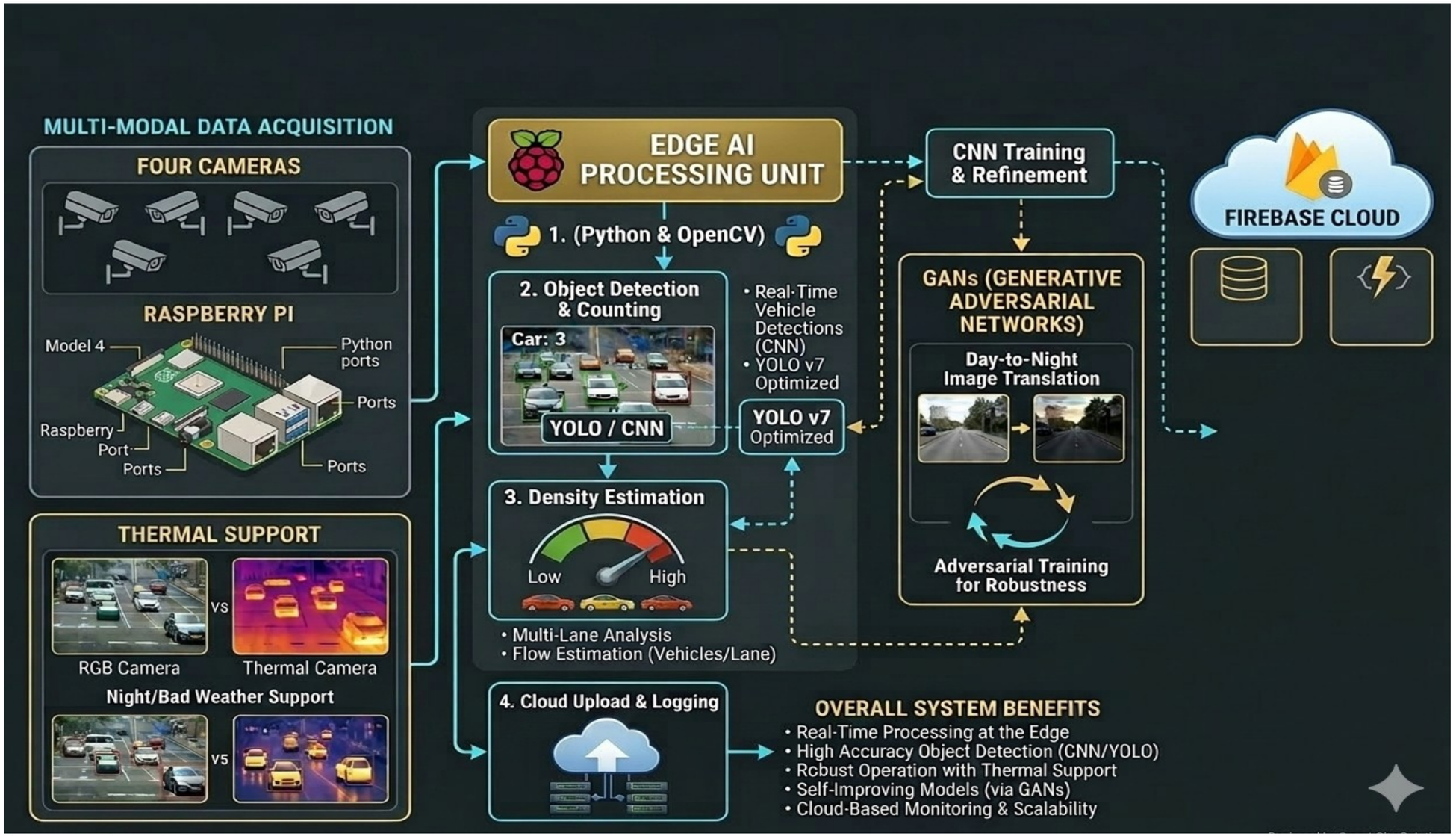
4. Traffic data is **safely sent to the cloud**
5. This makes the system **faster and more secure**

Ethics

6. System **does not store personal data**
7. **Faces and license plates are not recorded**



Implementation overview



IT22070494 |SIRIWARDANA W.L.N.S



Intelligent Traffic Light Control System Using GPS and Traffic Density Data

Bachelor of Science (Hons) in Information Technology Specializing in Information Technology

Problem Definition

- Many traffic lights still use fixed time control
- They cannot react properly to emergency situations
- Ambulances and fire brigade vehicles may be delayed at busy junctions
- Even if the signal changes, the emergency lane may still stay blocked
- Heavy traffic and slow driver reaction make the problem worse

Problem Solved

- My component solves this by using GPS data, ETA, route direction, and traffic density to change the traffic lights intelligently, clear the route early, activate a siren when needed, and keep the junction.

Intelligent Traffic Light Control

Goal

- Reduce emergency vehicle delay at a four-way junction using intelligent traffic light control.

Main purpose

- Use emergency vehicle GPS data, ETA, route direction, and live traffic density to change signal phases and clear the route early.

What it does

- Receives emergency vehicle data from the cloud
- Receives traffic density data from the camera module
- Analyzes urgency and congestion level
- Changes traffic light phases dynamically
- Starts preclearing before the emergency vehicle arrives
- Activates siren in Critical Mode when normal control is not enough
- Restores normal traffic flow after the emergency vehicle passes



Novelty

Key innovations

Our component introduces several important features that make it different from conventional traffic light control systems.

- **ETA Based Pre Clearing**
  Starts clearing the emergency route before the vehicle reaches the junction using distance, speed, and ETA.

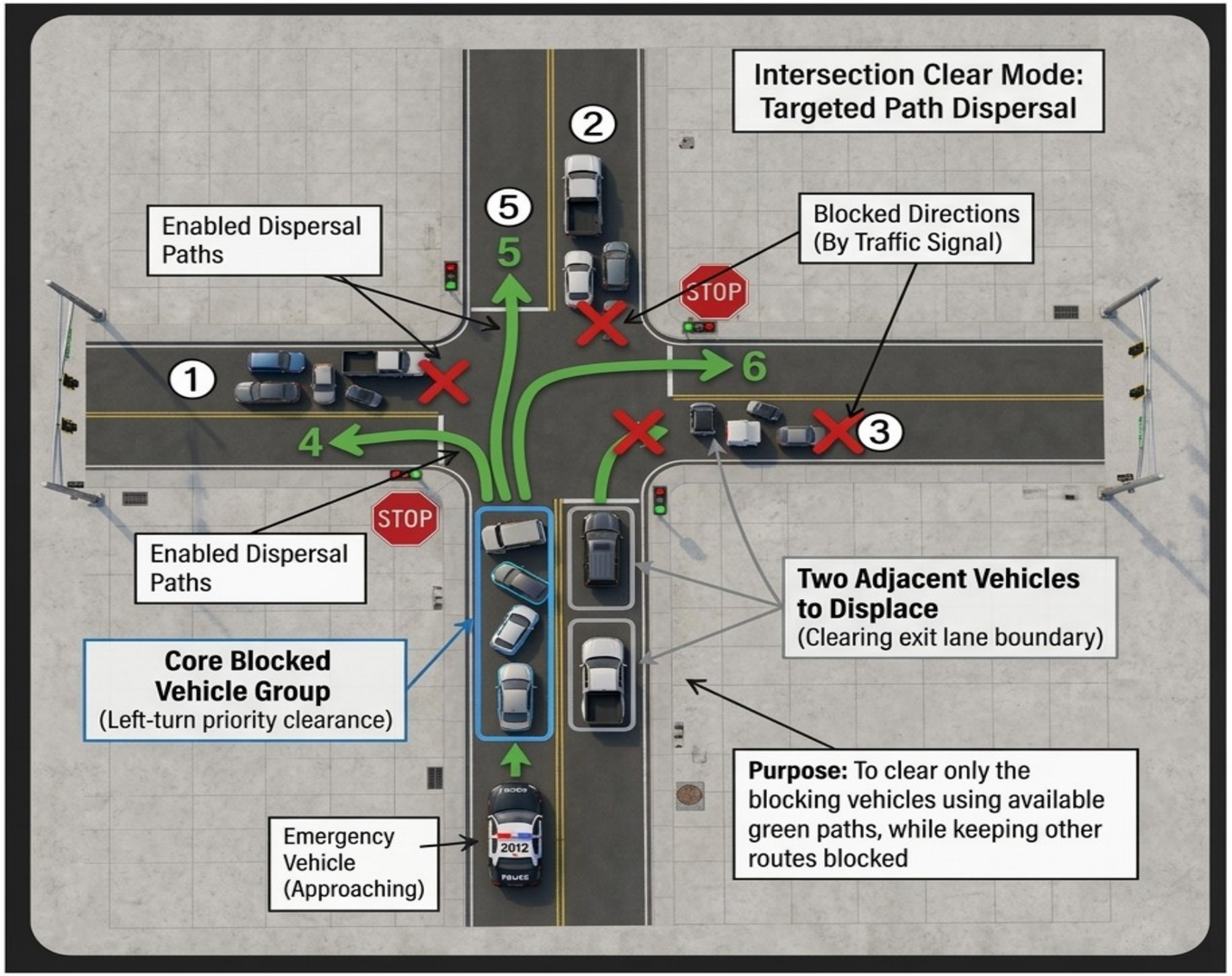
- **Traffic Density Based Signal Control**
  Adjusts signal phases using real time congestion levels instead of fixed time control.

- **Dynamic Signal Phase Selection**
  Decides which road gets green, how long it stays green, and when other roads must stop.

- **Siren Assisted Critical Clearance**
  Activates a siren when traffic is too dense and normal signal control cannot clear the emergency lane.

- **Emergency Lane Clearance Logic**
  Allows only the vehicles directly blocking the emergency vehicle to disperse through safe directions in Critical Mode.

Critical Clearance Mode



Research Objective

This component aims to:

Prioritize emergency vehicles at the junction.

Adjust signal timing dynamically.

Use both ETA and traffic density in decisions.

Restore normal traffic flow after emergency passage.

Activate a siren when traffic cannot be cleared by normal control.

Support fail-safe yellow blinking mode when abnormal conditions happen.

Methodology & Technologies

Technology Stack

- Microcontroller / NodeMCU
- Cloud Inputs (Firebase)
- Traffic Light Hardware
- Siren/Buzzer Module
- Embedded C++ / Python

Domain: Real-Time IoT & Embedded Systems

Intelligent Control Logic Workflow

1. Input

Data Acquisition

Receives real-time EV GPS, ETA, Route data & Density metrics from Cloud.

2. Process

Priority Rules

Calculates optimal phase timing based on ETA-based pre-clearing logic.

3. Action

Signal Control

Switches traffic lights to green for EV route; holds other lanes red.

4. Alert

Siren Trigger

Activates buzzer if traffic density remains critical despite green light.

Implementation Overview

Signal Control Logic Flow

Fetch Cloud EV Data + Density
Inputs: GPS, ETA, Vehicle Count

↓

Decision Engine Processing
Compare ETA vs Density vs Priority

↓

Execute Signal Command
Pre-clear Route or Trigger Siren

IoT Control Channel & Dependencies

- Low-Latency MQTT/HTTP
- Secured Commands
- NodeMCU / ESP32
- Fail-Safe Protocols

System Inputs & Outputs

- EV GPS / ETA
- Traffic Density
- Route Direction



Control Unit

- Signal Phases
- Siren / Buzzer
- Yellow Blinking

Validation, Security & Ethics

Validation

- Test whether the system reduces emergency vehicle delay
- Check whether the signal phase changes are correct
- Measure response time from data input to traffic light action
- Check siren activation timing in Critical Mode

Security

- Allow only authorized data sources and control access
- Protect communication between cloud, controller, and traffic light system
- Prevent unauthorized access or hacking attempts

Ethics

- Use the system only for authorized emergency vehicles
- Ensure safe prototype testing without creating road risk
- Avoid unsafe traffic control decisions during testing



IT22606242 | KITHSANDU E G H

Multi Emergency Vehicle Management at Four Way Junction



Contribution

- Handles multiple emergency vehicles at the same junction.
- Determines priority order dynamically.
- Supports safe traffic signal control.
- Reduces conflicts and delays.

Novelty

- Multi-vehicle prioritization system.
- Uses ETA, distance, traffic density, and direction.
- Dynamic safe sequencing logic.

Bachelor of Science (Hons) in Information Technology Specializing in Information Technology

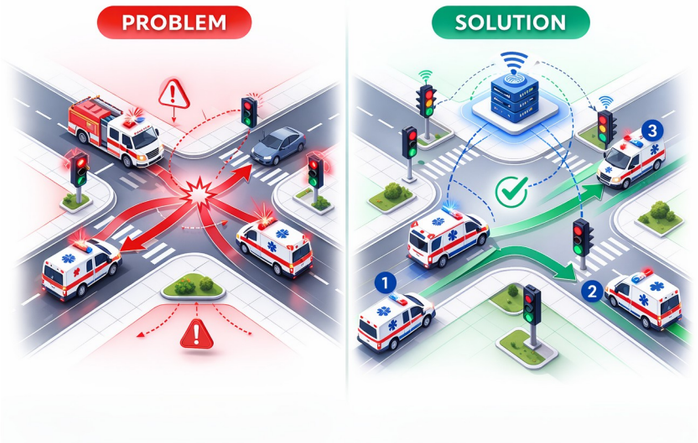
Problems Solved & Research Objectives

Problems Solved

- Multiple emergency vehicles arrive simultaneously
- Traditional systems prioritize only one vehicle
- Causes traffic conflicts and unsafe conditions

Research Objectives

- Detect multiple emergency vehicles
- Receive ETA and route data
- Identify route conflicts
- Compare urgency and direction
- Generate safe movement sequence



Made with GAMMA

Methodology & Technologies

Our system development is underpinned by a robust methodology and leverages cutting-edge technologies to ensure efficient and reliable emergency traffic management.

System Development & Evaluation

Iterative Approach

- **Design:** Comprehensive planning and architecture.
- **Develop:** Agile implementation of modules.
- **Test:** Rigorous validation and simulation.



Key Technologies Utilised

GPS Tracking

Precise vehicle location for real-time tracking.

Cloud Data Processing

High-speed data analysis and decision-making.

ETA Calculation

Dynamic prediction of arrival times for optimisation.

Smart Traffic Signals

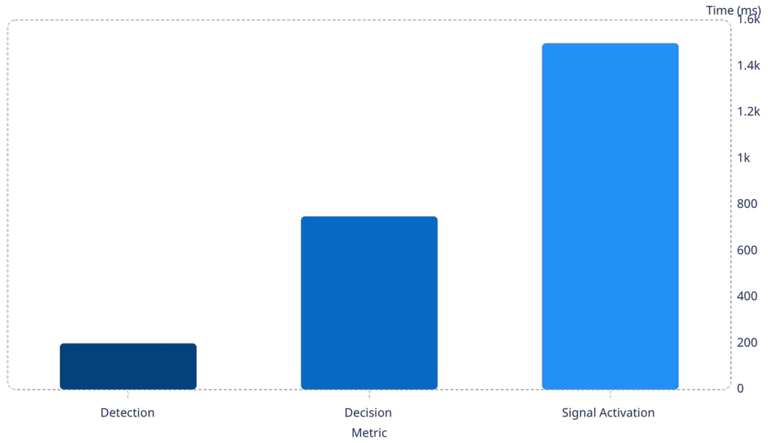
Responsive signal control for dynamic traffic flow.

Made with GAMMA

Validation, Security & Ethics

Performance Validation Metrics

- system response time
- High prioritisation accuracy
- Reduced emergency delay
- Fast conflict resolution



Average processing times for key system operations, illustrating efficiency gains.

Data Security Measures

Secure Data Transmission
GPS and ETA data are transmitted using encrypted channels.

Authorised Access Only
Access restricted to emergency services personnel and authenticated drivers.

Data Integrity
Advanced encryption ensures the integrity and confidentiality of all data.

Ethical Considerations

- Prioritising user privacy by anonymising non-essential data.
- Ensuring no unnecessary data storage or retention beyond operational needs.
- Limiting data usage strictly to emergency-related operational requirements.

Made with GAMMA

Overall System Architecture

Full integrated system

- GPS mobile application
- camera based traffic density detection
- cloud server
- single emergency vehicle priority management
- multiple emergency vehicle priority management
- intelligent traffic light control
- buzzer alert mechanism
- yellow blinking fail safe mode

Main message

All components work together as one complete smart emergency traffic management solution



Commercialization Plan

Financial Strategy, Revenue & IP Protection

Prototype Budget & Investment

Raspberry Pi 4 / 5 (8GB)	**LKR 80,000**
Google Maps APIs & Cloud	**LKR 110,000**
Cameras & Hardware	**LKR 43,750**
TOTAL ESTIMATED COST	**LKR 233,750**

Revenue Model Streams



Deployment

Initial setup & hardware installation fees

Licensing

Annual software subscription for municipalities

Maintenance

Ongoing technical support & updates

Customization

Tailored integration for specific city needs

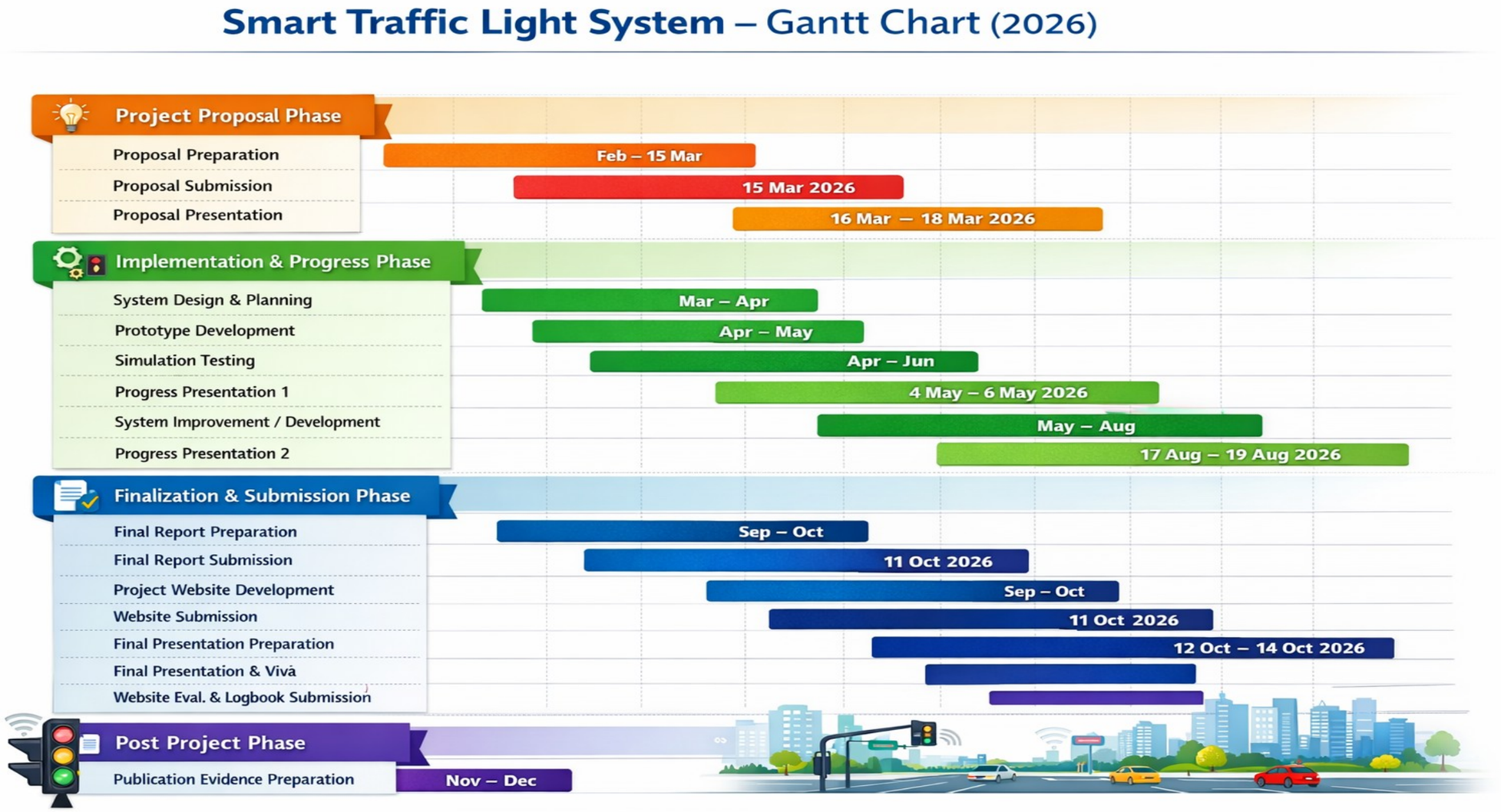
Pricing Strategy

- **Per-Junction Model:** Hardware + Software fee based on intersection count.
- **Fleet Subscription:** Tiered pricing based on number of registered emergency vehicles.
- **Service Tiers:** Basic (Traffic Control) vs. Premium (Analytics + Multi-EV Support).

Intellectual Property (IP)

- **Algorithm Protection:** Copyright for multi-vehicle conflict resolution & prioritization logic.
- **App Design:** UI/UX workflow protection for role-based emergency driver interface.
- **System Integration:** Proprietary method of linking GPS mobile data with edge-based traffic density analysis.

Gantt Chart


Smart Traffic Light System – Gantt Chart (2026)
Project Proposal Phase
Proposal Preparation
Feb – 15 Mar
Proposal Submission
15 Mar 2026
Proposal Presentation
16 Mar – 18 Mar 2026
Implementation & Progress Phase
System Design & Planning
Mar – Apr
Prototype Development
Apr – May
Simulation Testing
Apr – Jun
Progress Presentation 1
4 May – 6 May 2026
System Improvement / Development
May – Aug
Progress Presentation 2
17 Aug – 19 Aug 2026
Finalization & Submission Phase
Final Report Preparation
Sep – Oct
Final Report Submission
11 Oct 2026
Project Website Development
Sep – Oct
Website Submission
11 Oct 2026
Final Presentation Preparation
12 Oct – 14 Oct 2026
Final Presentation & Viva
Website Eval. & Logbook Submission
Post Project Phase
Publication Evidence Preparation
Nov – Dec

Work Breakdown Structure

Smart Emergency Traffic Light Control System

Work Breakdown & Development Timeline

Phase 1 – System Design & Requirements (Month 1–2)	Phase 2 – Core Module Development (Months 3–5)	Phase 3 – Core Module Development (Months 3–5)	Cloud Integration & Intelligent Control (Months 6–7)	Safety Mechanisms & Prototype Deployment (Month 8)	Phase 5 – Testing & Evaluation (Month 9)
Main Tasks	**Main Tasks**	**Main Tasks**		**Main Tasks**	**Outputs**
• Literature review and research analysis • System architecture design • Traffic junction hardware planning • Mobile app requirement design • Data collection planning for traffic monitoring	• Mobile app development for emergency drivers • GPS tracking integration for emergency vehicles • Camera-based traffic density detection system • Vehicle detection model training (Computer Vision) • Local Raspberry Pi processing setup	• Cloud data communication setup • Real-time GPS data processing • Traffic density data synchronization • Signal control logic development • Local Raspberry pi processing setup	• Cloud data communication setup • Real-time GPS data processing • Signal control logic development algorithm	• Siren warning mechanism implementation • Fail-sate traffic mode (yellow blinking safety mode) • Prototype setup at simulated four-way junction • Integration of all components	• System testing and validation • Performance evaluation • Emergency vehicle priority testing • System optimization and improvements • Documentation & final research preparation
Main Tasks	**Outputs**	**Outputs**	**Outputs**	**Outputs**	**Outputs**
• Finalized system architecture • Initial research proposal • Requirement specification	• Finalized system architecture • Initial research proposal • Requirement specification	• Mobile emergency control application • Camera-based traffic monitoring module	• Integrated smart traffic control system • Real-time decision making logic	• Complete working prototype • Emergency warning and safety system	• Final evaluation report • Validated smart traffic system • Final research documentation

THANK YOU!